Aranyabodh Annual Operating Cost Projection

November 2026 to October 2027

Revised 15 October 2026 | All amounts in INR | USD 1 = INR 100

Budget recommendation

Set aside INR 9.00 lakh for one year of operating costs at 1,000 registered users. Projected recurring cash requirements are INR 7.54 lakh, including the provisional tax allowance. Adding 15% contingency and INR 10,000 for initial provisioning gives a calculated requirement of INR 8.77 lakh.

Annual funding summary	INR
Recurring services and operating allowances before tax	638,866
Provisional tax allowance at 18%	114,996
Projected recurring cash requirement	753,862
Contingency at 15%	113,079
Initial provisioning allowance	10,000
Calculated first year funding requirement	876,941
Recommended annual allocation	900,000

Core assumptions

Planning input	Base assumption
Registered users	1,000 forest department officers and commercial users
Monthly active users	600 users
Average daily active users	200 users
Questions per active user daily	4.5 questions
Question volume	27,000 in a 30-day month and 328,500 over the year
Initial document library	500 documents averaging 20 pages each
Monthly document additions	100 documents averaging 20 pages each
Exchange rate	INR 100 per USD across all USD-priced items

Activity assumes intermittent use of a specialist work tool: 600 monthly active users average about 10 active days in a 30-day month. These are planning assumptions rather than measured usage or a validated industry benchmark. The forecast uses actual calendar days and holds the user base constant.

Scope: infrastructure, AI APIs, storage, backups, messaging, monitoring and business subscriptions. Development, maintenance labour, staff salaries, manual document review and sales labour are excluded. Costs are shown before subscription revenue.

1 Recurring cost breakdown

The comparison columns use a standard 30-day month. The annual column uses actual calendar days. All amounts exclude the provisional tax allowance. Fixed rupee envelopes remain unchanged; USD-priced hosting, database, Redis and AI are converted at INR 100 per USD.

Cost category	2026 month	2027 month	Year total
AI generation	23,100	42,158	474,166
Main app service including document processing	2,500	2,500	30,000
Supabase Pro and Small compute	3,000	3,000	36,000
Redis Cloud at USD 6 monthly	600	600	7,200
Storage backups and bandwidth envelope	1,000	1,000	12,000
Monitoring logging and error tracking	1,000	1,000	12,000
OCR and embedding envelope	500	500	6,000
Email SMS and WhatsApp envelope	2,000	2,000	24,000
Domain renewal allocation	125	125	1,500
Business and support subscriptions	2,000	2,000	24,000
Optional Jev provision	1,000	1,000	12,000
Static frontend DNS CDN and TLS	0	0	0
Operating subtotal	36,825	55,883	638,866

Hosting and document processing

One app service is budgeted at USD 25 per month. The API and document-processing process share the same paid VM or app-service allocation, whether run in one container or separate containers on that allocation. There is no separately billed document worker. This assumes spare CPU and memory in the main service; platform charges based on additional instances would require a revised estimate.

Supabase is USD 25 Pro plus USD 15 Small compute, less USD 10 compute credit, totalling USD 30 monthly. Redis Cloud is budgeted at the specified USD 6 monthly. The database provision assumes usage remains within the included disk and transfer limits.

Operating allowances

Retain INR 1,000 monthly for monitoring, logging, error tracking and uptime checks. Use included platform features first and spend only on gaps or overages. This is a total envelope, not an additional fee for each monitoring feature.

Storage assumes approximately 100 GB including originals and backup copies. Communications allow roughly 1,000 SMS, 4,000 WhatsApp utility messages and 6,000 emails monthly, including provider fees. The INR 2,000 business-tool envelope covers modest email, billing and support subscriptions. These are budget caps, not confirmed vendor invoices.

Jev remains optional and off by default in the code. Its INR 1,000 provision covers about INR 907 at 27,000 questions if decision inputs average 8,000 tokens per question at USD 0.042 per million. No Gemini savings are assumed. If unused, the provision is not spent.

2 AI and document processing assumptions

A user question can involve rewriting, planning, answer generation and verification. The generation estimate aggregates those stages and billable reasoning tokens. The January increase reflects the previously verified change to Gemini Flash pricing; all other provider rates are held constant over the projection.

Model USD per million tokens	Input	Output
Gemini 3.5 Flash Lite throughout the year	0.30	2.50
Gemini 3.8 Flash through December 2026	0.75	3.75
Gemini 3.8 Flash from January 2027	1.50	7.50

Question class	Share	2026 INR	2027 INR
Simple	25%	0.1111	0.1111
Procedural	50%	0.6667	1.2000
Complex	25%	1.6667	3.1667

Class costs are allowances before recovery attempts, converted from the earlier INR 90 per USD basis by multiplying by 100 ÷ 90. Procedural questions assume Flash contributes 80% of their generation bill; complex questions assume 90%. Those components double in January. Simple questions use Flash Lite. These assumptions are consistent with the project architecture and selected cost evaluations but require validation against live traffic.

Calculation method

Generation cost per question = weighted class cost × 1.10. The 10% uplift covers recovery attempts and is applied once. The resulting planning rates are INR 0.85556 through December 2026 and INR 1.56139 from January 2027, before tax.

For a 30-day month: 200 daily active users × 4.5 questions × 30 days = 27,000 questions. Generation costs are approximately INR 23,100 through December and INR 42,158 from January. No caching discounts, batch discounts or promotional credits are assumed.

OCR and embeddings

Assume half of each month’s 2,000 new pages require OCR. For text indexing, allow 600 tokens per page and 20% chunk overlap, giving 1.44 million document tokens monthly. Query embeddings allow 200 aggregate input tokens per question.

At INR 100 per USD, the indicative 30-day monthly cost is INR 150 for OCR, INR 29 for document embeddings and INR 108 for query embeddings. Their combined INR 287 sits within the INR 500 envelope. This uses USD 1.50 per 1,000 OCR pages and USD 0.20 per million text-embedding tokens, conservatively ignoring OCR free allowances.

Tax and contingency

Apply an 18% provisional tax allowance to the operating subtotal without deducting potential input credits. Actual tax treatment follows supplier invoices. Add 15% contingency to the resulting recurring cash requirement. This reserve covers moderate exchange-rate movement, overages and estimation uncertainty; it is not an expected invoice charge.

3 Monthly cash and funding projection

Period: 1 November 2026 to 31 October 2027. Other recurring costs total INR 13,725 monthly. Cash need includes the provisional tax allowance and excludes contingency and initial provisioning. All monetary columns are INR.

Month	Questions	AI cost	Other costs	Tax	Cash need
Nov 2026	27,000	23,100	13,725	6,629	43,454
Dec 2026	27,900	23,870	13,725	6,767	44,362
Jan 2027	27,900	43,563	13,725	10,312	67,600
Feb 2027	25,200	39,347	13,725	9,553	62,625
Mar 2027	27,900	43,563	13,725	10,312	67,600
Apr 2027	27,000	42,158	13,725	10,059	65,941
May 2027	27,900	43,563	13,725	10,312	67,600
Jun 2027	27,000	42,158	13,725	10,059	65,941
Jul 2027	27,900	43,563	13,725	10,312	67,600
Aug 2027	27,900	43,563	13,725	10,312	67,600
Sep 2027	27,000	42,158	13,725	10,059	65,941
Oct 2027	27,900	43,563	13,725	10,312	67,600
Year	328,500	474,166	164,700	114,996	753,862

Funding allocation

Period	Monthly allocation	Period total
November 2026 including initial allowance	65,000	65,000
December 2026	55,000	55,000
January to October 2027 each month	78,000	780,000
Annual allocation		900,000

The annual allocation covers INR 753,862 recurring cash need, INR 113,079 contingency and INR 10,000 initial provisioning, leaving INR 23,059 of additional headroom. Average recurring cash need is INR 62,822 per month.

Calendar length explains monthly variation within each price period. The larger January step comes from the Flash price increase. Calculations use full precision; displayed figures are rounded to the nearest rupee, so displayed row sums may differ slightly from totals.

4 Scenarios and budget boundaries

The scenarios vary daily active users while holding the registered population at 1,000, questions at 4.5 per active day and other recurring envelopes unchanged. Higher use may also require a larger main app allocation.

Scenario	DAU	Recurring cash per year	With contingency and initial costs
Low	100	474,104	555,220
Base	200	753,862	876,941
High	350	1,173,499	1,359,524

Initial provisioning allowance

One time provision	INR
Initial document OCR and embeddings including tax allowance	1,100
Conditional sender registration or messaging onboarding	6,000
Other initial vendor fees and provisioning headroom	2,900
Total initial allowance	10,000

For the assumed 10,000-page initial library, 50% OCR share and chunk overlap, OCR and embeddings are about INR 894 before tax or INR 1,055 with the tax allowance. The other initial amounts are conditional provisions, not quoted or committed fees. The domain is already counted in the recurring allocation.

Costs outside this operating scope

Development and maintenance labour, employee and contractor salaries, manual document review, sales and marketing labour are excluded. This is an app operating budget, not a forecast of total company expenditure.

Payment processing depends on receipts. If online subscriptions are collected, provision approximately 3% of gross receipts as a separate allowance and replace it with the selected payment arrangement. Revenue is not assumed or deducted in this report.

Paid content licences, formal security audits, enterprise support, dedicated or restricted hosting, contractual availability guarantees, advanced document extraction, web search, voice and imagery are not assumed. Add them only if the product or procurement scope requires them. Managed point-in-time database recovery is also outside the current backup envelope.

When to revise the budget

Check question volume, generation cost, database size, transfer, logs and messaging monthly. Review capacity when shared document processing affects answer latency, and revise the forecast when monthly questions exceed 30,000 or generation cost per question exceeds the allowance by 20%. Use actual invoices to replace provisional envelopes; included services and unused provisions should reduce spend.